JONAS SCHULT

Senior AI Research Scientist at Meta · London, UK

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SUMMARY

Senior AI Research Scientist at Meta in London, working on generative image and video models from research to production. PhD research in 3D scene understanding at RWTH Aachen. Author of Mask3D, a standard baseline for 3D instance segmentation, with two ScanNet benchmark wins and publications at CVPR, ICCV, ICLR, ICRA and 3DV, including orals at CVPR and 3DV.

EXPERIENCE

Senior AI Research Scientist Aug 2024 – present
Meta London, UK

Generative AI for image and video advertising. Promoted to Senior AI Research Scientist in March 2026.

- Lead the automated quality evaluation of generated images and videos: set the technical direction, run the evaluation systems in production, mentor team members, and build the annotation and tooling behind them.
- Built the team's image-generation serving platform from scratch, including monitoring and on-call. It is deployed as part of Meta's generative AI tools for advertisers.
- Developed and launched new video-generation recipes after validating them in online experiments, and coordinated the launches across teams.
- Built reward models for prompt following in video generation.
- Owned a background-generation model end to end, from the data pipeline to deployment at scale.

Research Scientist Intern, 3D Generative AI Jun 2023 – Nov 2023
Meta GenAI Menlo Park, CA, USA

- ControlRoom3D: 3D room generation by steering 2D diffusion models with semantic proxy rooms (CVPR 2024).

Doctoral Researcher, Computer Vision Chair (Prof. Bastian Leibe) Feb 2020 – Jul 2024
RWTH Aachen University Aachen, Germany

- Research on 3D scene understanding for point clouds, meshes and LiDAR sequences, with 10+ publications at venues including CVPR, ICCV, ICLR, ICRA and 3DV.
- Supervised six master's students, leading to three conference papers. Teaching assistant for Operating Systems and Advanced Computer Vision: managed 24 student TAs and taught exercise classes for 800+ students.

Student Research Assistant, 3D Scene Understanding Sep 2017 – Sep 2019
Computer Vision Chair, RWTH Aachen University Aachen, Germany

- Deep learning for semantic and instance segmentation of 3D point clouds; first paper at ECCV 2018 Workshops.

Research Intern, Computer Vision Sep 2016 – Dec 2016
Volkswagen Group Research Wolfsburg, Germany

- 2D object and keypoint detection for automated driving.

EDUCATION

PhD Candidate in 3D Scene Understanding Feb 2020 – Jul 2024
RWTH Aachen University, advisor: Prof. Bastian Leibe Aachen, Germany

- Dissertation *Understanding and Generating 3D Scenes* written; defense pending.

MSc Computer Science, grade 1.2 (with distinction) Oct 2017 – Jan 2020
RWTH Aachen University Aachen, Germany

- Master's thesis (grade 1.0) on DualConvMesh-Net, published as a CVPR 2020 Oral. Specialization in machine learning and computer vision, minor in Economics, Erasmus semester at NTNU Trondheim, Norway (spring 2017).

BSc Computer Science, grade 1.6 Oct 2013 – Sep 2016
RWTH Aachen University Aachen, Germany

- Bachelor's thesis (grade 1.0): *Clustering of Attributed and Evolving Graphs*. Minor in Electrical Engineering.

German grade scale: 1.0 = best, 4.0 = pass.

SELECTED FIRST-AUTHOR PUBLICATIONS

Full list on Google Scholar: 1,530+ citations (October 2026). * shared first authorship.

Mask3D: Mask Transformer for 3D Semantic Instance Segmentation
J. Schult, F. Engelmann, A. Hermans, O. Litany, S. Tang, B. Leibe
ICRA 2023. First mask transformer for 3D instance segmentation and 1st place on the ScanNet benchmark. 590+ citations and 700+ GitHub stars; a standard baseline in the field.

Mix3D: Out-of-Context Data Augmentation for 3D Scenes
A. Nekrasov*, **J. Schult***, O. Litany, B. Leibe, F. Engelmann
3DV 2021 (Oral). 1st place on the ScanNet semantic segmentation benchmark; 215+ citations.

DualConvMesh-Net: Joint Geodesic and Euclidean Convolutions on 3D Meshes
J. Schult*, F. Engelmann*, T. Kontogianni, B. Leibe
CVPR 2020 (Oral). Learns jointly on meshes and point clouds with geodesic and Euclidean convolutions. Master's thesis work; 115+ citations.

ControlRoom3D: Room Generation using Semantic Proxy Rooms
J. Schult, S. Tsai, L. Höllein, B. Wu, J. Wang, C.-Y. Ma, K. Li, X. Wang, F. Wimbauer, Z. He, P. Zhang, B. Leibe, P. Vajda, J. Hou
CVPR 2024. Generative 3D by steering 2D latent diffusion models with semantic proxy rooms (Meta GenAI internship); 80+ citations.

3D Segmentation of Humans in Point Clouds with Synthetic Data
A. Takmaz*, **J. Schult***, I. Kaftan, M. Akçay, B. Leibe, R. Sumner, F. Engelmann, S. Tang
ICCV 2023. Synthetic-data training for human-centric 3D scene understanding, in collaboration with ETH Zurich.

Point2Vec for Self-Supervised Representation Learning on Point Clouds
K. Knaebel*, **J. Schult***, A. Hermans, B. Leibe
GCPR 2023. Self-supervised pre-training for point clouds; 80+ citations.

FURTHER PUBLICATIONS

TUNA: Taming Unified Visual Representations for Native Unified Multimodal Models
Z. Liu et al. (incl. **J. Schult**). *CVPR 2026 (Highlight).*

HiStream: Efficient High-Resolution Video Generation via Redundancy-Eliminated Streaming
H. Qiu et al. (incl. **J. Schult**). *CVPR 2026.*

Tuna-2: Pixel Embeddings Beat Vision Encoders for Multimodal Understanding and Generation
Z. Liu et al. (incl. **J. Schult**). *arXiv preprint, 2026.*

AGILE3D: Attention Guided Interactive Multi-object 3D Segmentation
Y. Yue, S. Mahadevan, **J. Schult**, F. Engelmann, B. Leibe, K. Schindler, T. Kontogianni. *ICLR 2024.*

Mask4Former: Mask Transformer for 4D Panoptic Segmentation
K. Yilmaz, **J. Schult**, A. Nekrasov, B. Leibe. *ICRA 2024.*

M2F3D: Mask2Former for 3D Instance Segmentation
J. Schult, A. Hermans, F. Engelmann, O. Hilliges, S. Tang, B. Leibe. *CVPR 2022 Workshops (Spotlight).*

Know What Your Neighbors Do: 3D Semantic Segmentation of Point Clouds
F. Engelmann, T. Kontogianni, **J. Schult**, B. Leibe. *ECCV 2018 Workshops.*

AWARDS & DISTINCTIONS

- 1st place, ScanNet 3D instance segmentation benchmark (Mask3D, at time of publication).
- 1st place, ScanNet 3D semantic segmentation benchmark (Mix3D, at time of publication).
- 2nd place, Urban3D instance segmentation challenge, ECCV 2022 (Mask3D).
- Dean's List, RWTH Aachen (top 5% of students), 2018–2019.
- Deutschlandstipendium, 2017–2019 • Erasmus+ scholarship, 2017.

ACADEMIC SERVICE

- Reviewer: CVPR 2023, 2024, 2026 • ICCV 2023 • WACV 2023.
- Co-organizer of the OpenSUN3D workshop series on open-vocabulary 3D scene understanding at ICCV 2023, CVPR 2024 and ECCV 2024.

SKILLS

ML & methods: 3D deep learning (point clouds, meshes, LiDAR), transformers, diffusion models, self-supervised learning, reward modeling, LLM/VLM-based evaluation

Engineering: production ML in Python/PyTorch (model serving, monitoring, on-call), inference optimization, data pipelines and annotation operations, online experimentation (A/B tests)